OXFORD AQA
INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

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Forename(s) ____________________

Candidate signature ____________________
I declare this is my own work.

INTERNATIONAL GCSE
BIOLOGY
Paper 1

Thursday 22 May 2025 07:00 GMT Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice
In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
TOTAL	

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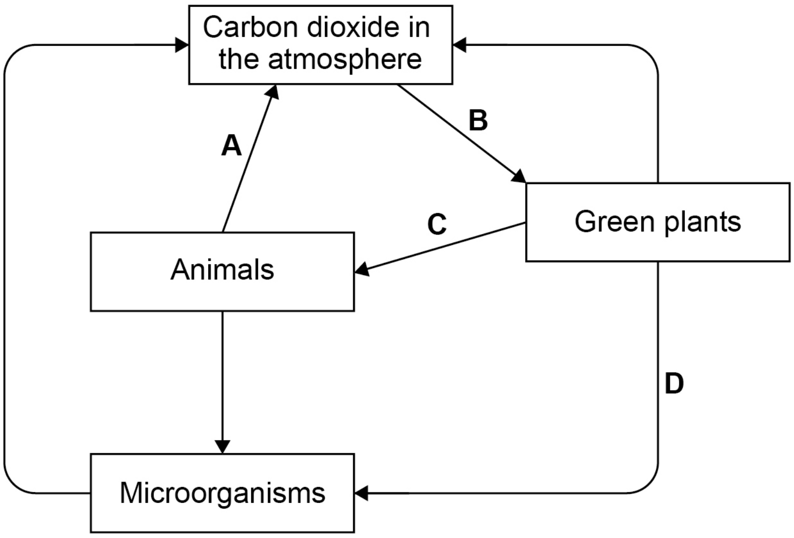
Answer **all** questions in the spaces provided.

0 1 Carbon is continually cycled between living organisms and the environment.

Figure 1 shows part of the carbon cycle.

Figure 1



0 1 . 1 Draw **one** line from each process to the letter that shows the process.

[3 marks]

Process	Letter in Figure 1
	A
Feeding	B
Photosynthesis	C
Respiration	D

Do not write outside the box

0 1 . 2 Carbon dioxide is released when fossil fuels are burnt.

Fossil fuels are burnt to heat houses.

Give **one other** material that releases carbon dioxide when burnt to heat houses.

[1 mark]

____

0 1 . 3 The concentration of carbon dioxide in the atmosphere has increased over the last 100 years.

Give **one** effect of the increased concentration of carbon dioxide in the atmosphere.

[1 mark]

____

____

0 1 . 4 Decay is part of the carbon cycle.

Which organisms are responsible for decay?

[1 mark]

Tick (✓) **one** box.

Bacteria ☐

Birds ☐

Trees ☐

Question 1 continues on the next page

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0 1 . 5 Bread will decay when it is left for many days.

Keeping bread in certain conditions can slow down the rate of decay.

Table 1 gives information about bread decay.

Complete **Table 1**.

[2 marks]

Table 1

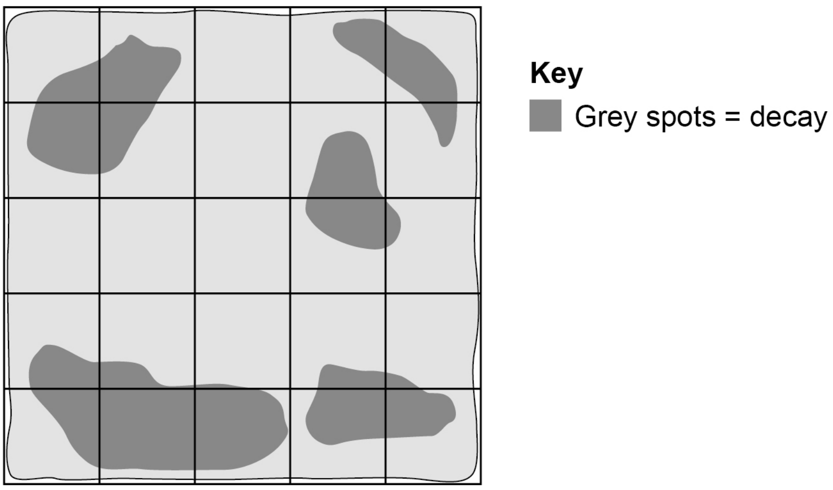
How bread is kept	Reason for slower decay
In a container with no air	
	Enzymes are less active

A piece of bread was left to decay.

As the bread decays grey spots develop on the bread.

Figure 2 shows the piece of bread.

Figure 2



0 1 . 6 Estimate the number of full squares of bread that show decay.

[1 mark]

Number of squares = ____________________

Do not write outside the box

0 1 . 7 Calculate the percentage of bread that shows decay.

Use your answer from Question **01.6**.

[2 marks]

Percentage of bread that shows decay = ____________ %

0 1 . 8 Five students calculated the percentage of decay on a different piece of bread.

The results were:

72% 92% 68% 76% 72%

Draw a ring around the anomalous result.

[1 mark]

12

Turn over for the next question

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There are no questions printed on this page

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ANSWER IN THE SPACES PROVIDED

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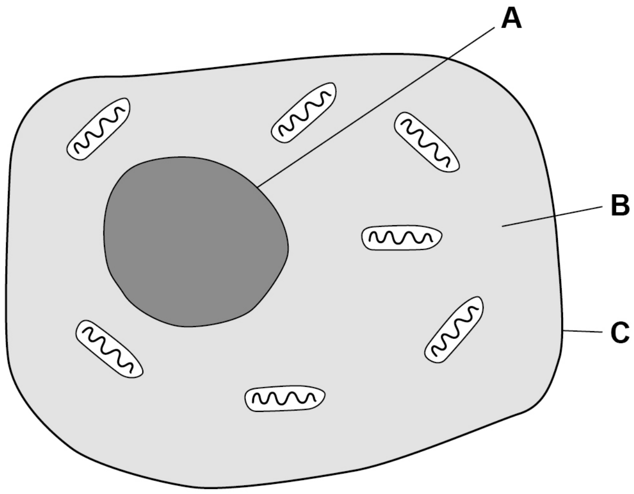
0 2 Some characteristics of organisms are controlled by DNA.

0 2 . 1 **Figure 3** shows a human cell.

Figure 3



Which part of the cell contains DNA?

[1 mark]

Tick (✓) **one** box.

A ☐ **B** ☐ **C** ☐

0 2 . 2 DNA codes for amino acids.

Complete the sentences.

[2 marks]

Choose answers from the box.

bases	**carbohydrates**	**lipids**	**proteins**

DNA contains four different compounds called ________________________ .

Amino acids are joined together to make ________________________ .

Turn over ►

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Offspring inherit DNA from their parents.

Scientists have studied inheritance for over 100 years.

0 2 . 3 One of the first people to study inheritance worked with pea plants.

What was the name of the person who studied inheritance in pea plants?

[1 mark]

____

A tall pea plant was crossed with a short pea plant.

All the offspring were tall pea plants.

Height in pea plants is controlled by two alleles:

T = allele for tall plants

t = allele for short plants.

0 2 . 4 Which term describes the allele for tall plants?

[1 mark]

Tick (✓) **one** box.

Dominant ☐

Phenotype ☐

Recessive ☐

Do not write outside the box

0 2 . 5 A different tall pea plant was crossed with a short pea plant.

Complete **Figure 4** to show the genotypes of the offspring.

[2 marks]

Figure 4

Tall pea plant

		T	**t**
Short pea plant	**t**	Tt	
	t		

0 2 . 6 Draw a ring around the genotype of any offspring that would be a tall pea plant.

[1 mark]

0 2 . 7 Two pea plants with the genotype **TT** were grown in a garden.

The pea plants did **not** grow to the same height.

Suggest **one** reason why the plants did not grow to the same height.

Do **not** refer to genes in your answer.

[1 mark]

____

____

Question 2 continues on the next page

Turn over ►

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0 2 . 8 Alleles are passed on to new cells during cell division.

Cells can divide by mitosis or by meiosis.

Complete **Table 2** to show features of mitosis and meiosis.

[2 marks]

Tick (✓) **one** box in each row.

Table 2

	Mitosis	**Meiosis**
Produces four cells		
Produces genetically identical cells		
Produces gametes		

0 2 . 9 Tumours develop when cell division is uncontrolled.

Draw **one** line from each type of tumour to the description of the tumour.

[2 marks]

Type of tumour	**Description**
Benign tumours	can differentiate to form any type of cell
Malignant tumours	do **not** invade other tissues
	form secondary tumours

13

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0 3 Pathogens are microorganisms that cause disease.

0 3 . 1 Some bacteria are pathogens.

What do bacteria produce that make us feel ill?

[1 mark]

Tick (✓) **one** box.

Bile ☐

Hormones ☐

Toxins ☐

0 3 . 2 Antibiotics kill bacteria.

Name **one** antibiotic.

[1 mark]

____

0 3 . 3 Why are doctors advised **not** to overuse antibiotics?

[1 mark]

____

____

Question 3 continues on the next page

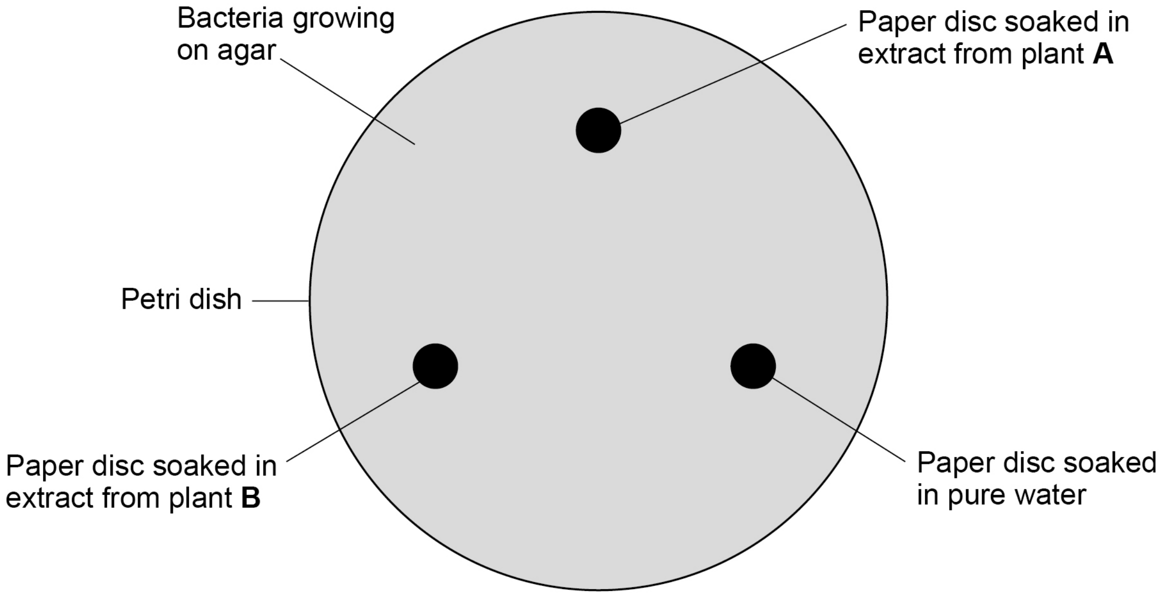
Do not write outside the box

Some plants can be used to kill bacteria.

A student investigated how effective two plants were at killing bacteria.

Figure 5 shows the apparatus used in the investigation.

Figure 5



This is the method used.

1. Crush leaves of plant **A** in water to make an extract.
2. Crush leaves of plant **B** in water to make an extract.
3. Soak paper discs in the extracts from the two plants.
4. Soak another paper disc in pure water.
5. Put the paper discs onto bacteria growing on agar as shown in **Figure 5**.
6. Leave the Petri dish at 25°C for 48 hours.
7. Measure the diameter of any clear zone around each paper disc.

Do not write outside the box

0 3 . 4 **Table 3** shows a risk assessment for the investigation.

Complete **Table 3**.

[2 marks]

Table 3

Hazard	**Risk**	**How to minimise risk**
Bacteria	Infection from the bacteria	
Glass Petri dish	Cut from broken glass	

0 3 . 5 Explain why a paper disc soaked in pure water was used in the investigation.

[2 marks]

Question 3 continues on the next page

Figure 6 and **Table 4** show the results.

Figure 6

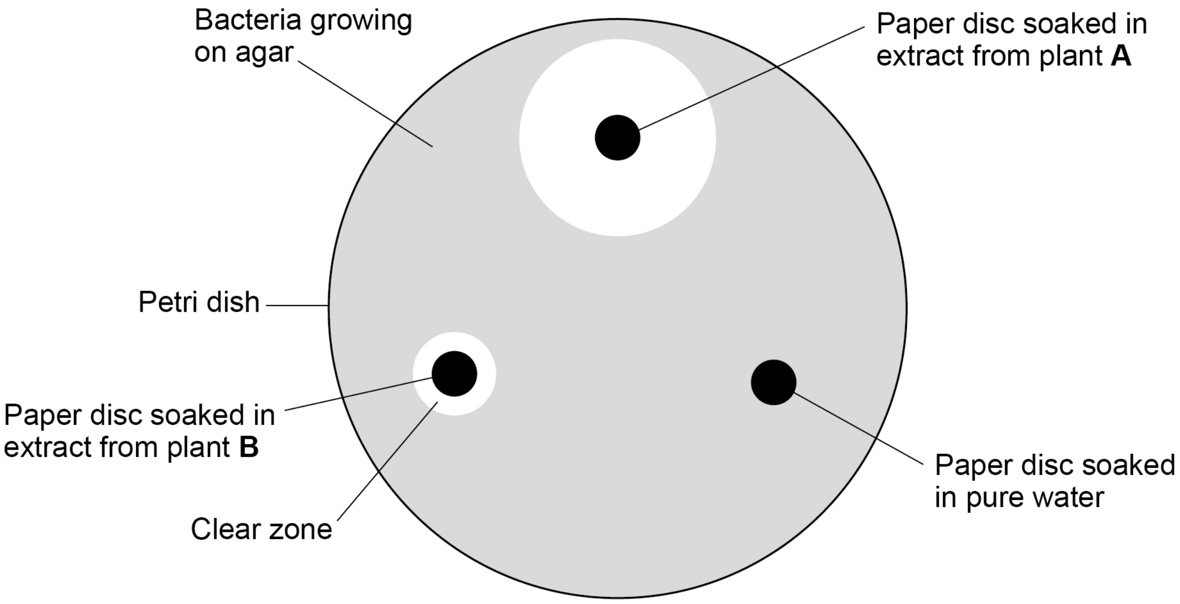


Table 4

Paper disc	**Area of clear zone in mm^2**
Extract from plant **A**	**X**
Extract from plant **B**	95
Pure water	0

Do not write outside the box

0 3 . 6 Calculate value **X** in **Table 4**.

Use the equation:

$$\text{area of a circle} = \pi r^2$$

r = radius of circle

$\pi = 3.14$

[4 marks]

X = ______________________ mm^2

The student concluded:

'Plant **A** is more effective at killing all bacteria than plant **B**.'

0 3 . 7 Explain the evidence for the student's conclusion.

[2 marks]

Question 3 continues on the next page

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0 3 . 8 Give **one** reason why the student's conclusion may **not** be correct.

Do **not** refer to repeating the investigation in your answer.

[1 mark]

0 3 . 9 Insects transfer pathogens to plants when they feed on the plants.

Lectin is a protein which can kill insects.

Cotton plants can be genetically engineered to produce lectin.

Explain why it might be an advantage **to farmers** to grow the genetically engineered cotton plants.

[3 marks]

17

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Turn over for the next question

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ANSWER IN THE SPACES PROVIDED

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0 4 Some birds migrate to different parts of the world during winter.

When birds migrate they fly long distances.

0 4 . 1 Suggest why some birds need to migrate during winter.

[1 mark]

____

____

0 4 . 2 Why do birds need a lot of energy to fly long distances?

[1 mark]

____

____

0 4 . 3 Energy is released during respiration.

Complete the word equation for aerobic respiration.

[2 marks]

glucose + ____________ $\longrightarrow$ ____________ + ____________

Do not write outside the box

0 4 . 4 **Table 5** shows the energy content of some types of food.

Table 5

Type of food	Energy content in kJ/g
Carbohydrate	17
Fat	37
Protein	17

The body mass of a bird increases by more than 100% before migration.

Most of the increase in body mass is caused by increased stores of fat.

Explain why birds store fat rather than carbohydrate or protein.

[2 marks]

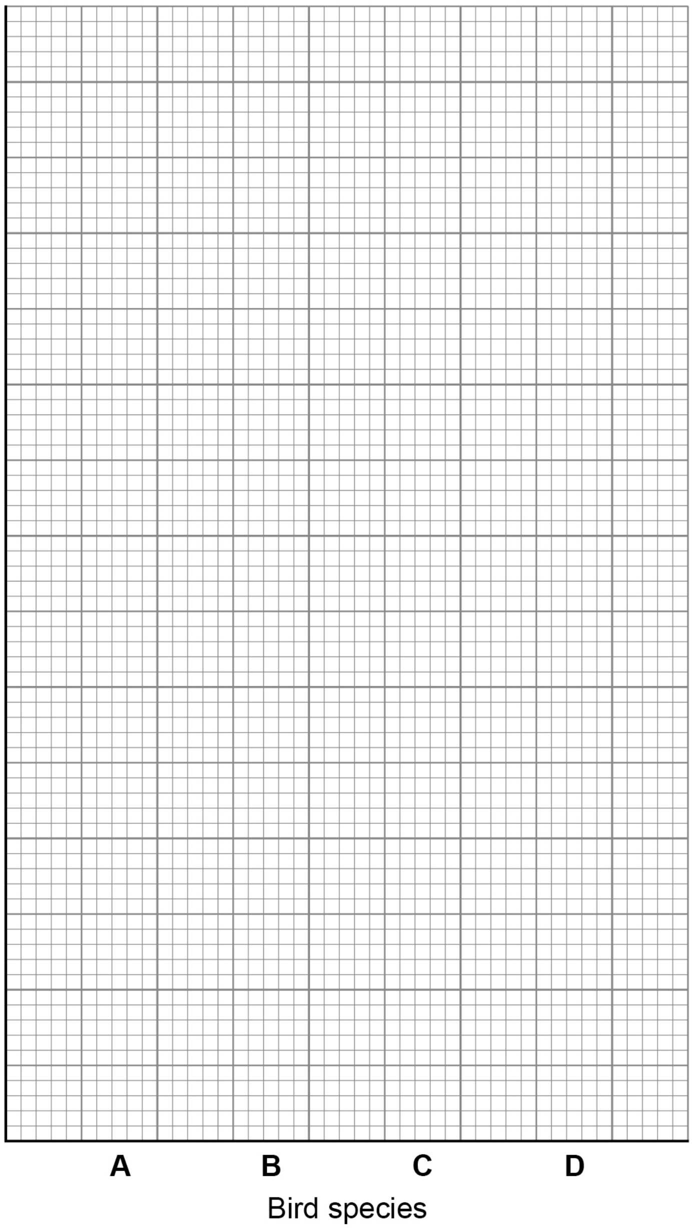
Question 4 continues on the next page

Table 6 shows the distance some birds travel during migration.

Table 6

Bird species	Distance travelled during migration (km)
A	64 000
B	18 500
C	24 000
D	8 000

Figure 7



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0 4 . 5 Complete the bar graph in **Figure 7**.

You should:

- label the y axis
- add a suitable scale to the y axis
- plot the data from **Table 6**.

[3 marks]

0 4 . 6 Calculate the percentage increase of the distance travelled by species **A** compared to species **D**.

[3 marks]

Percentage increase = ____________ %

Question 4 continues on the next page

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0 4 . 7 Two species of birds have evolved by natural selection from a common ancestor.

Describe how evolution occurs by natural selection.

[3 marks]

0 4 . 8 Scientists discovered two birds on an island which were similar in appearance.

One of the birds was male and the other bird was female.

Describe how the scientists could show that the birds were two members of the same species.

[1 mark]

16

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0 5 Waste materials are removed from the human body.

0 5 . 1 Which organ **system** removes carbon dioxide from the body?

[1 mark]

0 5 . 2 The kidneys remove waste materials from the body.

The waste materials leave the body in urine.

Describe how a healthy kidney produces urine.

[5 marks]

Question 5 continues on the next page

When a kidney stops working this is called kidney failure.

A person with kidney failure can be given:

- a kidney transplant
- dialysis treatment.

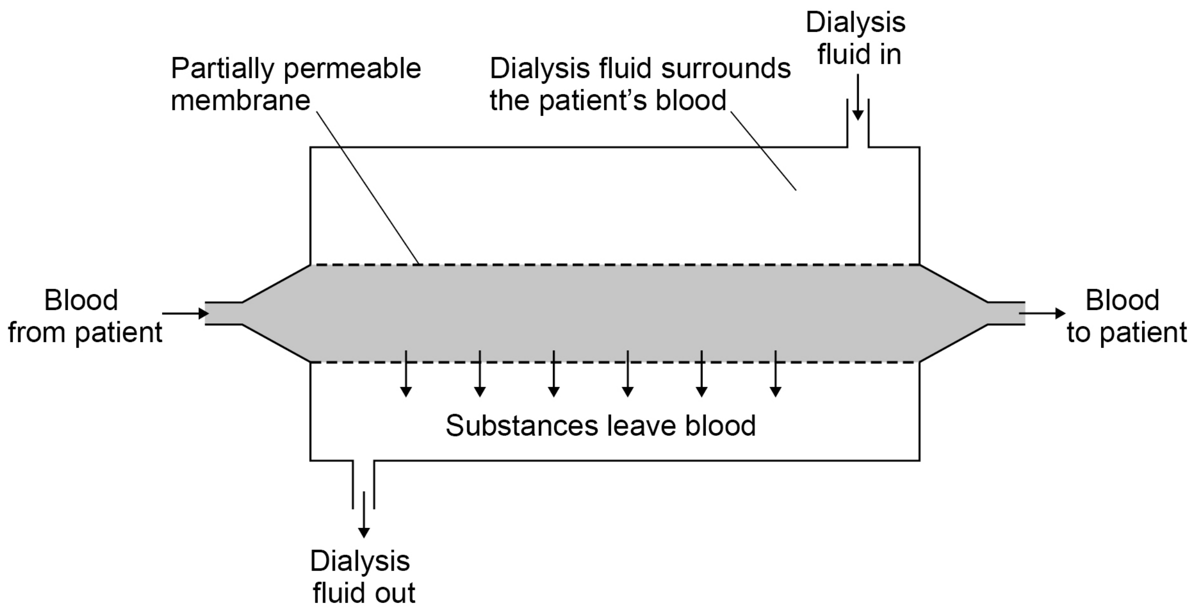
A kidney transplant needs a kidney from a donor.

Dialysis treatment involves three dialysis sessions every week.

A dialysis session lasts about 8 hours.

Figure 8 shows how dialysis works.

Figure 8



0 5 . 3 Suggest **one** advantage of using **dialysis** to treat a person with kidney failure compared with a kidney transplant.

[1 mark]

Do not write outside the box

0 5 . 4 Suggest **one** advantage of using a **kidney transplant** to treat a person with kidney failure compared with dialysis.

[1 mark]

Question 5 continues on the next page

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0 5 . 5 The concentrations of substances in dialysis fluid need to be carefully controlled.

Table 7 shows the concentrations of different substances in blood and in two possible dialysis fluids.

Table 7

	Concentration in arbitrary units			
	Normal range in blood	**Patient's blood**	**Fluid A**	**Fluid B**
Glucose	80 to 100	100	70	100
Sodium ions	135 to 140	140	140	140
Potassium ions	2 to 3	5	3	2
Urea	1 to 4	30	5	0

Explain why fluid **B** should be used as dialysis fluid for treating the patient rather than fluid **A**.

[6 marks]

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Extra space

0 5 . 6 Explain why the production of excess ADH results in a low concentration of sodium ions in the blood.

[2 marks]

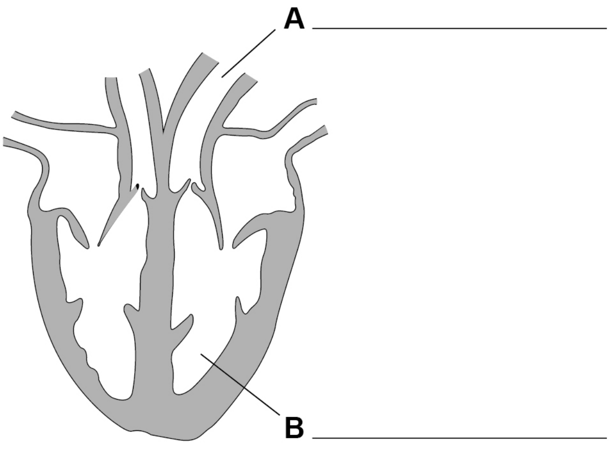
16

Turn over for the next question

0 6 . 1 The heart is an organ that pumps blood around the body.

Figure 9 shows a human heart.

Figure 9



Label **A** and **B** in **Figure 9**.

[2 marks]

0 6 . 2 Blood vessels transport blood around the body.

Explain why the structure of arteries is different from the structure of veins.

[4 marks]

____

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Capillaries are blood vessels that connect arteries to veins.

Dengue fever is a disease that affects capillaries.

In severe cases of dengue fever the capillaries break open and blood passes out of the capillaries.

0 6 . 3 Describe the process of blood clotting to prevent blood loss from the capillaries.

[4 marks]

Question 6 continues on the next page

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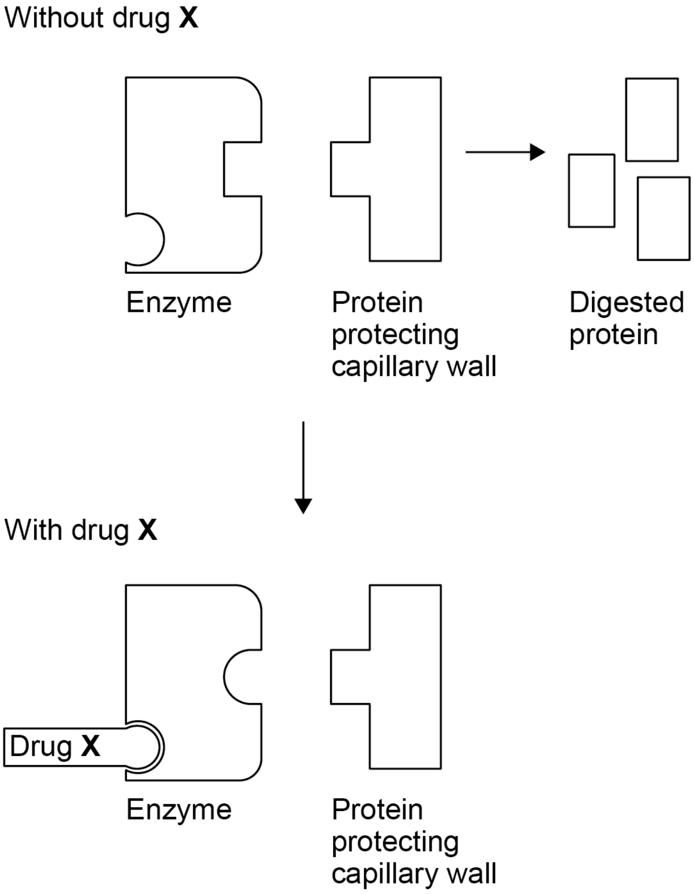
0 6 . 4 The dengue fever virus causes cells to make an enzyme.

The enzyme digests proteins that normally prevent capillary walls from breaking.

Drug X is given to people infected with dengue fever.

Figure 10 shows a model of how **drug X** works.

Figure 10



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Explain how **drug X** reduces bleeding in people infected with dengue fever.

[4 marks]

Question 6 continues on the next page

0 6 . 5 In a severe case of blood loss, a patient needs a blood transfusion.

It is important that a compatible blood type is given to the patient.

Table 8 shows a blood compatibility table.

Table 8

		Donor blood type			
		A	**B**	**AB**	**O**
Recipient blood type	**A**	✓			✓
	B		✓		✓
	AB	✓	✓	✓	✓
	O				✓

The patient has type **A** blood.

Only blood from donors of type **A** and type **O** can be given safely to the patient.

Explain why.

[2 marks]

16

END OF QUESTIONS

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